1 A spring is attached to a strong cylindrical magnet of length L and cross-sectional area A. The magnet is placed on thin card on top of a magnetic sheet on the bench, as shown in Fig. 1.1.

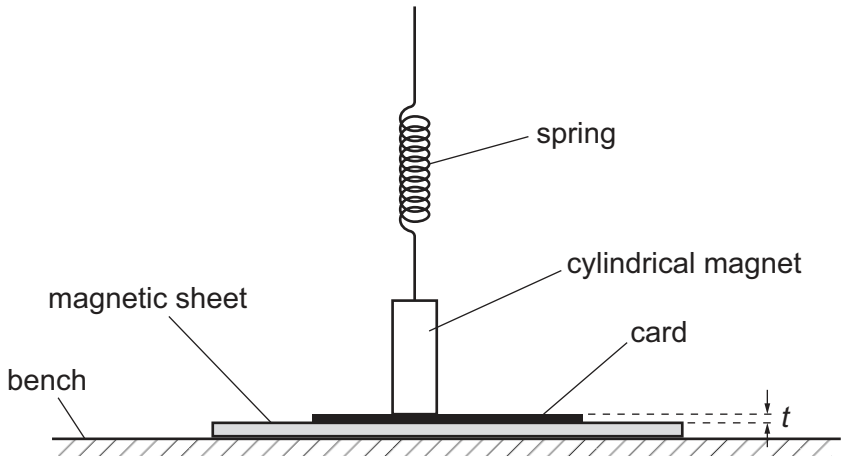

Fig. 1.1

The thickness of the card is t. The magnetic flux density at one of the poles of the magnet is B.

A force is applied upwards to the spring. The extension of the spring when the magnet just leaves the card is s.

It is suggested that s is related to t by the relationship

$$ks = \frac{ALBZ}{t}$$

where k is the spring constant of the spring and Z is a constant.

Plan a laboratory experiment to test the relationship between s and t.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for Z.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

https://xtremepape.rs/